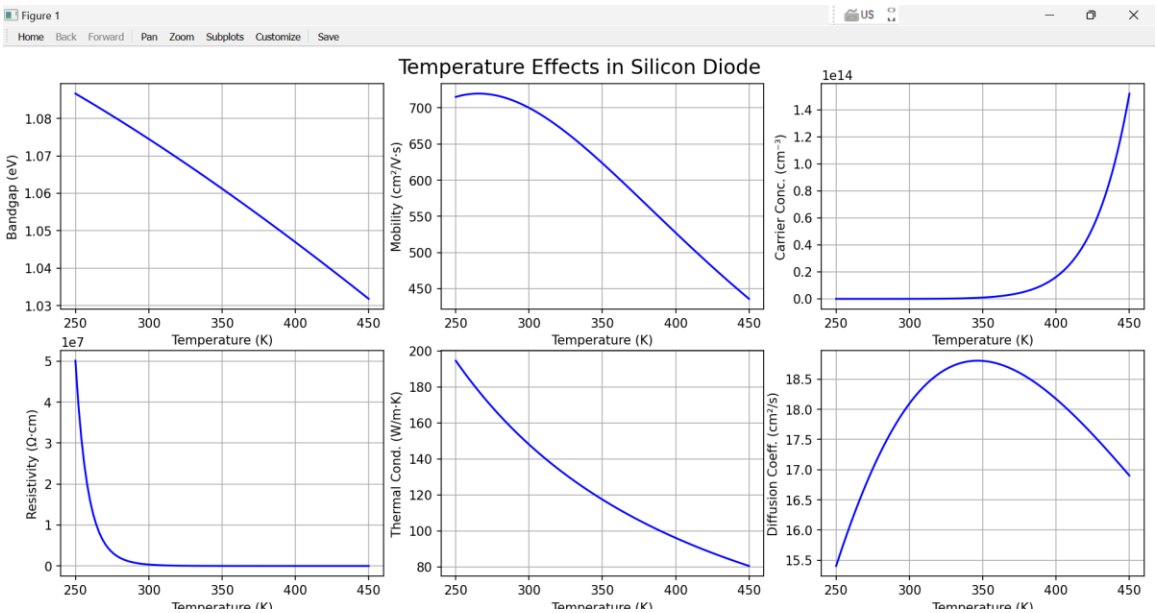
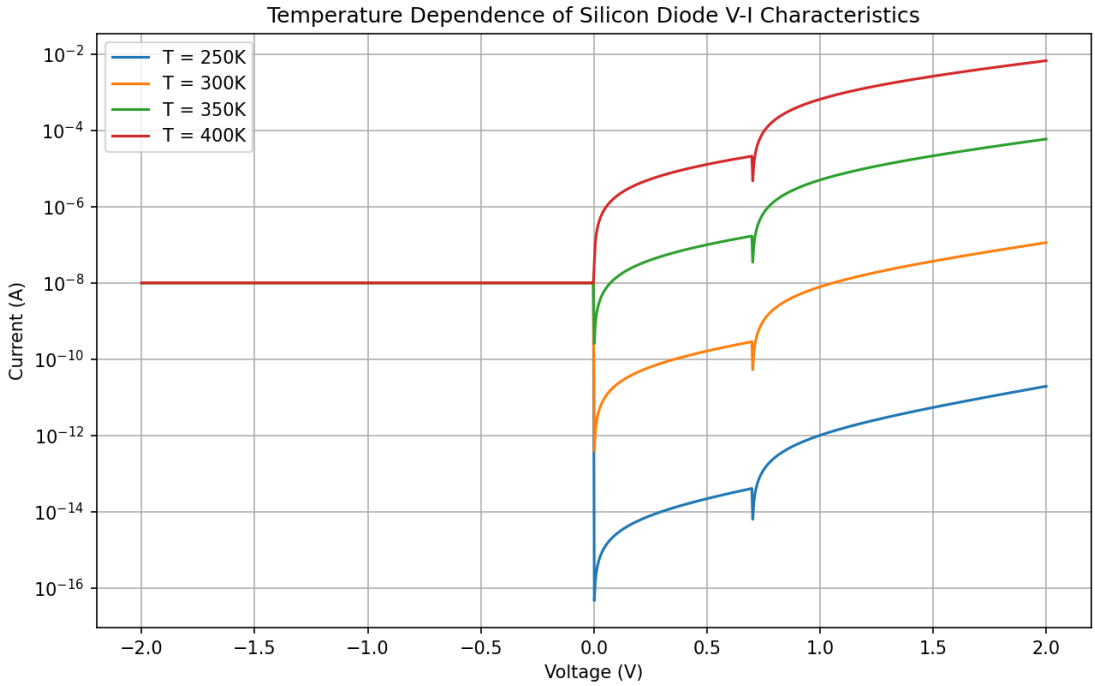


Figure 1
Home Back Forward Pan Zoom Subplots Customize Save
Temperature Effects in Silicon Diode
Bandgap (eV)
Mobility (cm²/V·s)
Carrier Conc. (cm⁻³)
1e14
Resistivity (Ω·cm)
1e7
Thermal Cond. (W/m·K)
Diffusion Coeff. (cm²/s)
Temperature (K)


Temperature Dependence of Silicon Diode V-I Characteristics
T = 250K
T = 300K
T = 350K
T = 400K
Current (A)
Voltage (V)

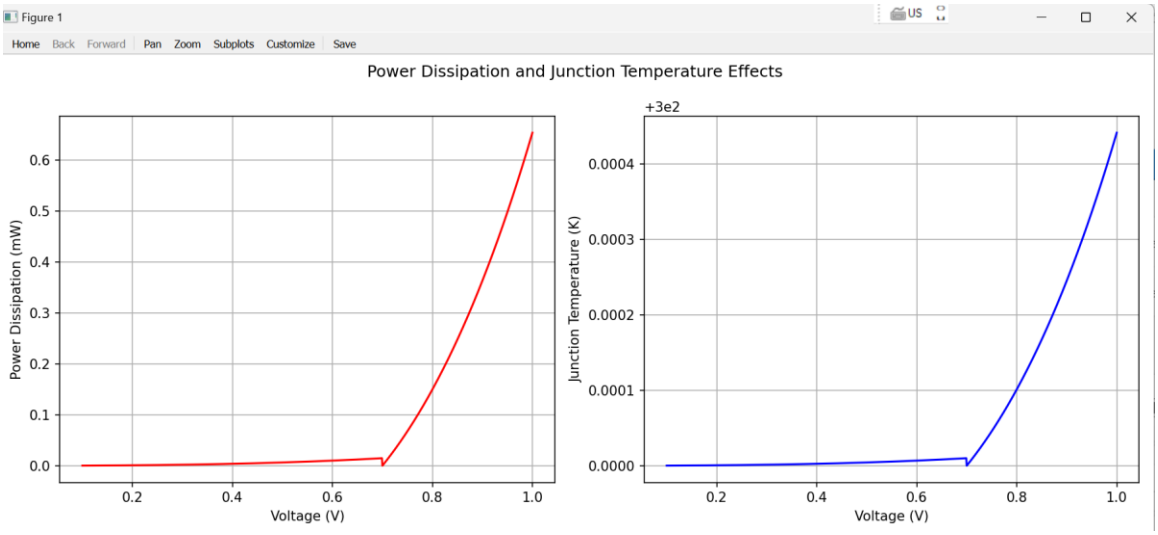
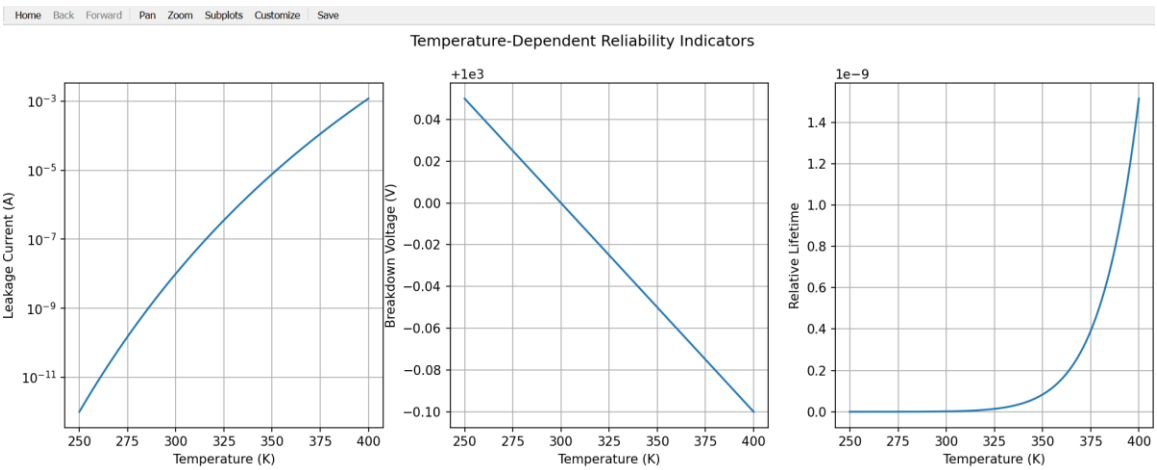
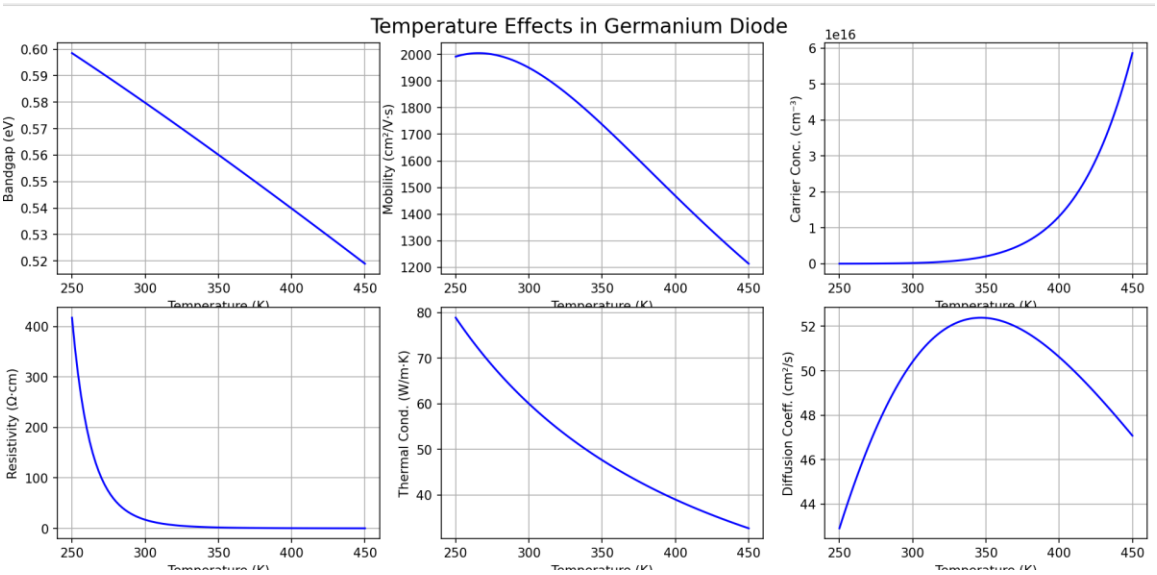
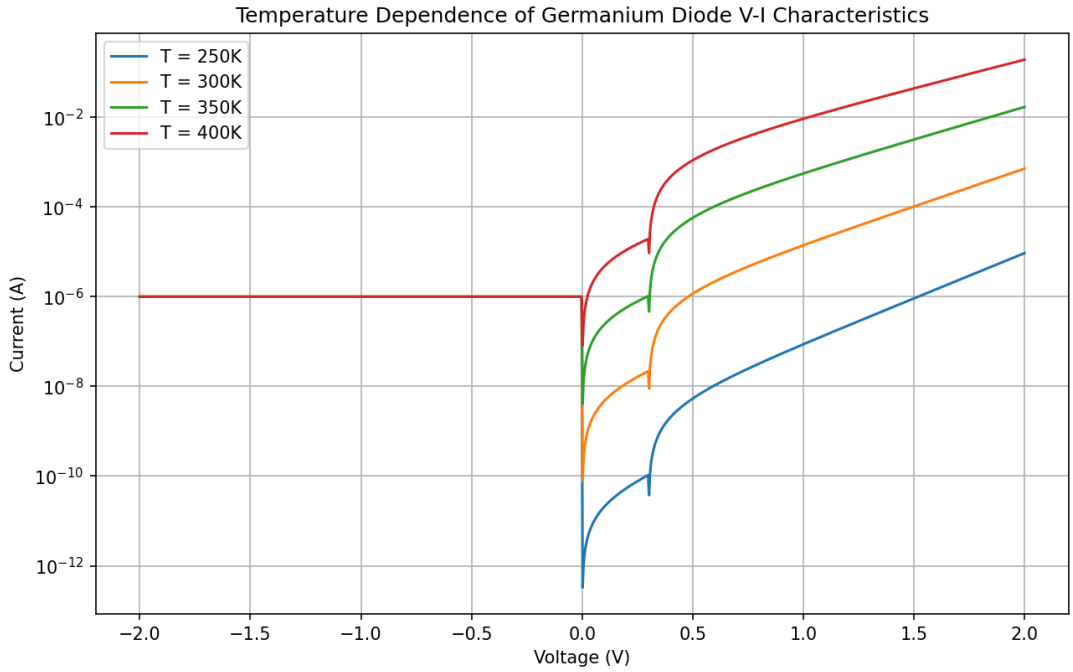
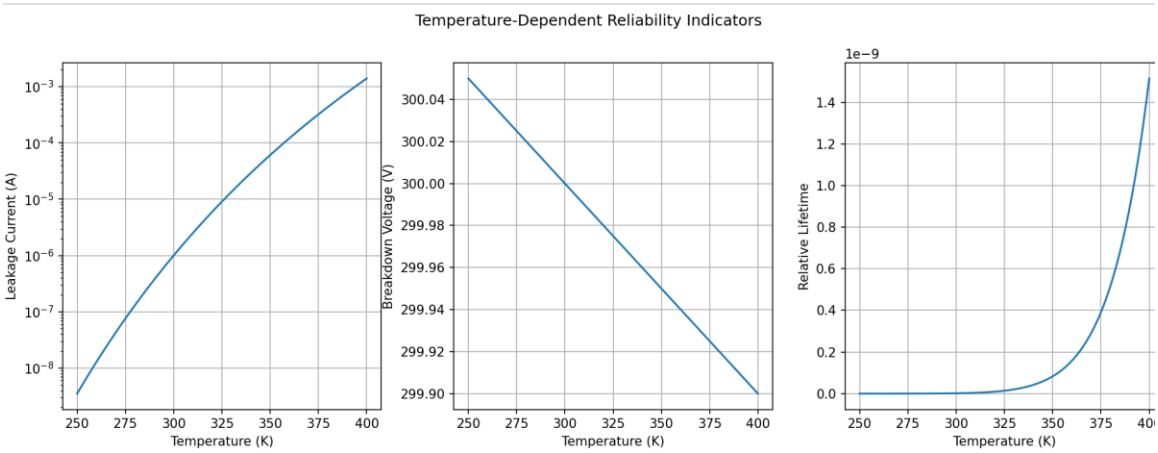


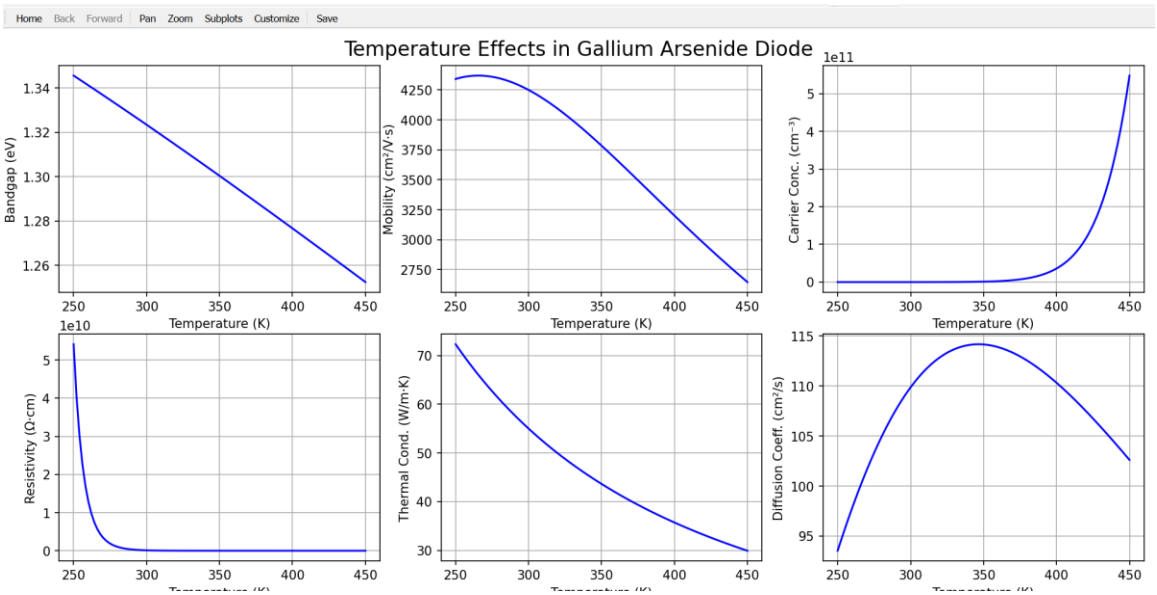
Figure 1
Home Back Forward Pan Zoom Subplots Customize Save
Power Dissipation and Junction Temperature Effects
Power Dissipation (mW)
Voltage (V)
+3e2
Junction Temperature (K)
Voltage (V)

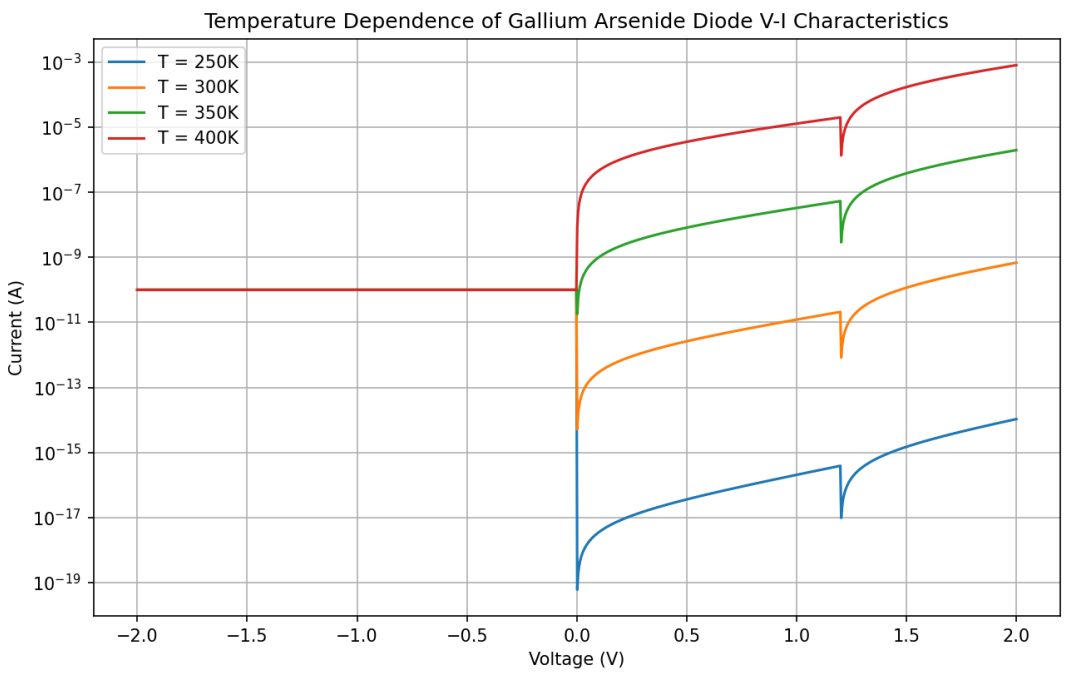

Home Back Forward Pan Zoom Subplots Customize Save
Temperature-Dependent Reliability Indicators
Leakage Current (A)
Temperature (K)
+1e3
Breakdown Voltage (V)
Temperature (K)
1e−9
Relative Lifetime
Temperature (K)

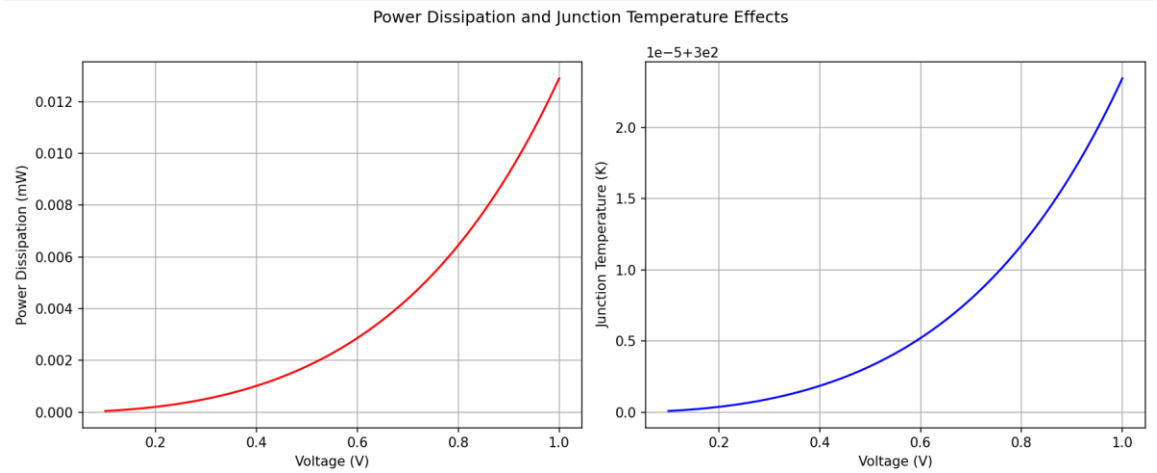

Temperature Effects in Germanium Diode
Bandgap (eV)
Mobility (cm²/V·s)
1e16
Carrier Conc. (cm⁻³)
Resistivity (Ω·cm)
Thermal Cond. (W/m·K)
Diffusion Coeff. (cm²/s)

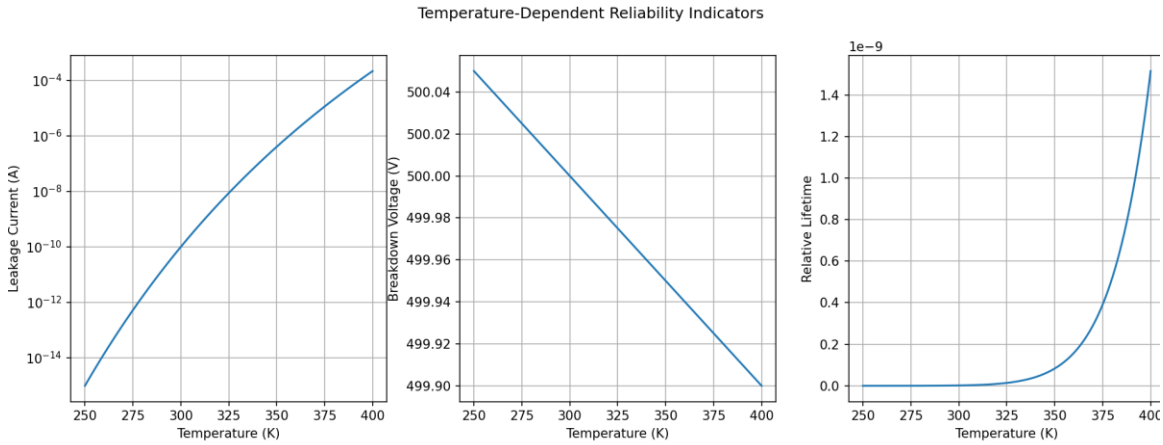

Temperature Dependence of Germanium Diode V-I Characteristics
T = 250K
T = 300K
T = 350K
T = 400K
Current (A)
Voltage (V)


Temperature-Dependent Reliability Indicators
Leakage Current (A)
Breakdown Voltage (V)
Relative Lifetime
1e−9
Temperature (K)


Home Back Forward Pan Zoom Subplots Customize Save
Temperature Effects in Gallium Arsenide Diode
Bandgap (eV)
Mobility (cm²/V·s)
Carrier Conc. (cm⁻³)
1e11
Resistivity (Ω·cm)
1e10
Thermal Cond. (W/m·K)
Diffusion Coeff. (cm²/s)
Temperature (K)


Temperature Dependence of Gallium Arsenide Diode V-I Characteristics
T = 250K
T = 300K
T = 350K
T = 400K
Current (A)
Voltage (V)


Power Dissipation and Junction Temperature Effects
Power Dissipation (mW)
Voltage (V)
1e−5+3e2
Junction Temperature (K)
Voltage (V)


Temperature-Dependent Reliability Indicators
Leakage Current (A)
Temperature (K)
Breakdown Voltage (V)
Temperature (K)
1e−9
Relative Lifetime
Temperature (K)